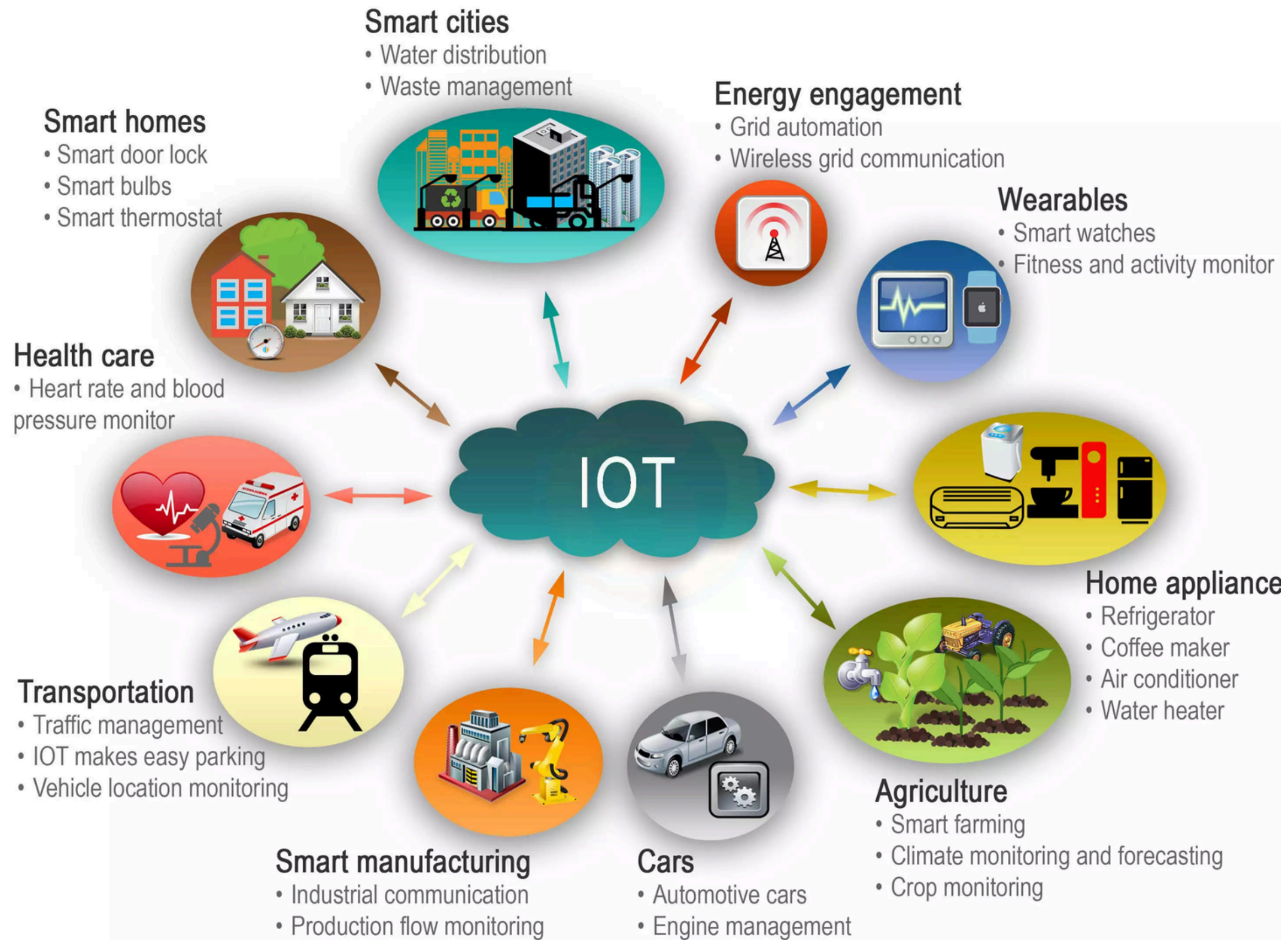


HOME AUTOMATION

IOT

Team members

- 1.S.Yellareddy AP23110011648
- 2.K.Sainath naik AP23110011646
- 3.M.Geethendra naik AP23110011652
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- 5.S. Naresh naik AP23110011651

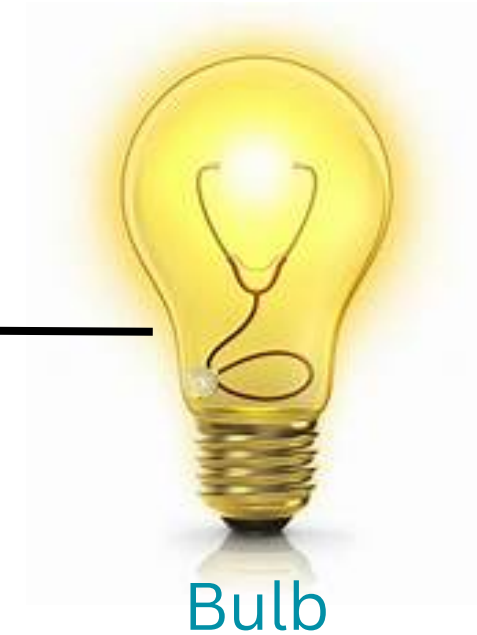
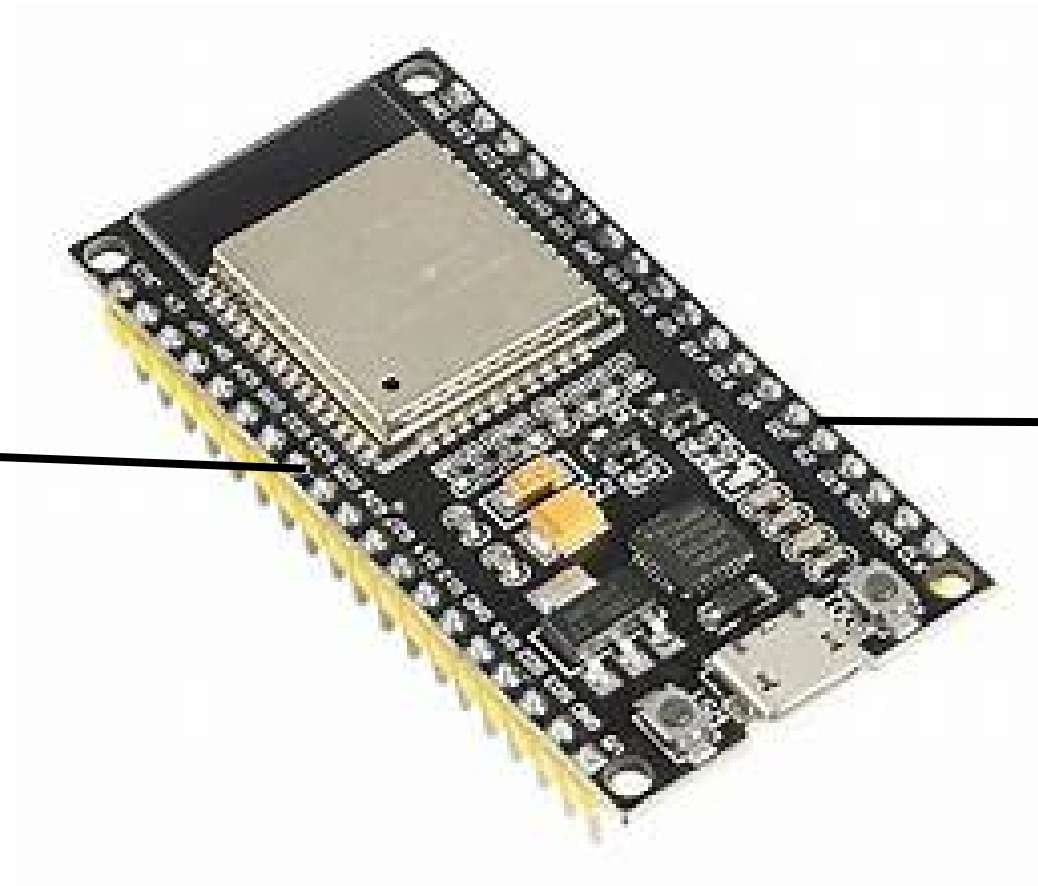


How ESP32 controll home appliances

Home Automation



Power supply

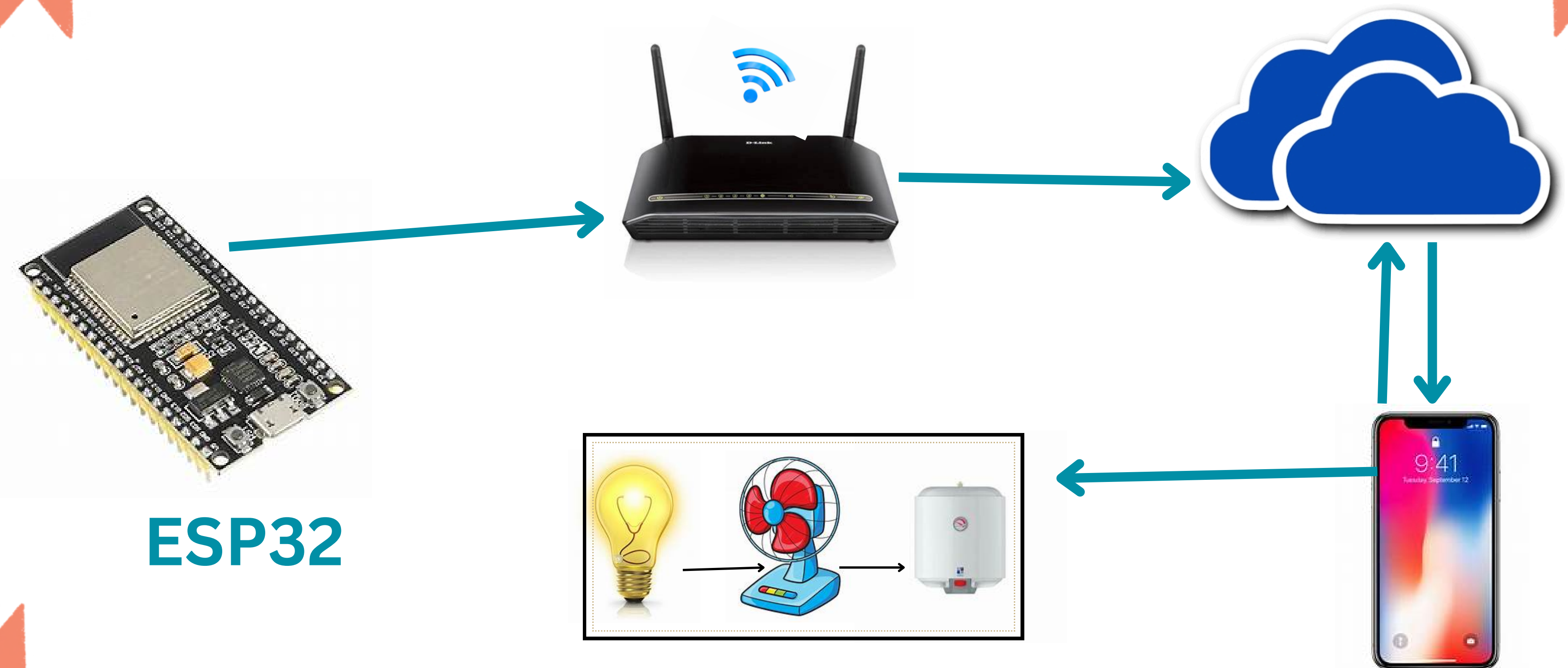


Bulb

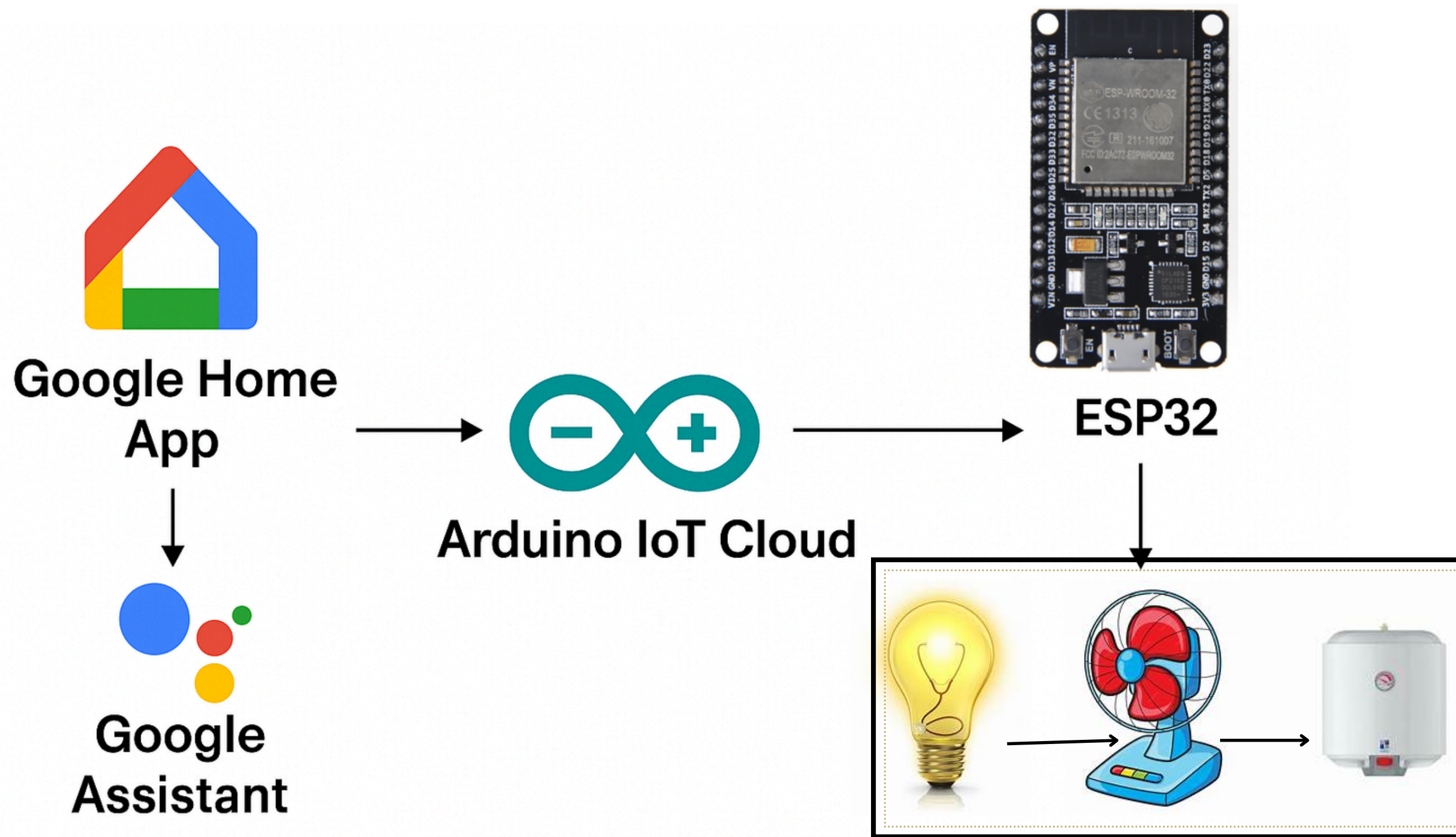


OFF :cuts the power supply
ON :connects the power supply

Controlling Devices Via cloud

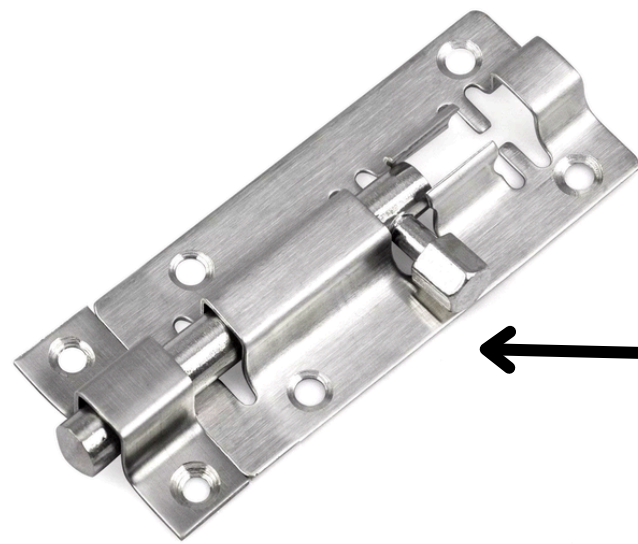


How ESP32 Connects to Google Home via Arduino IoT Cloud

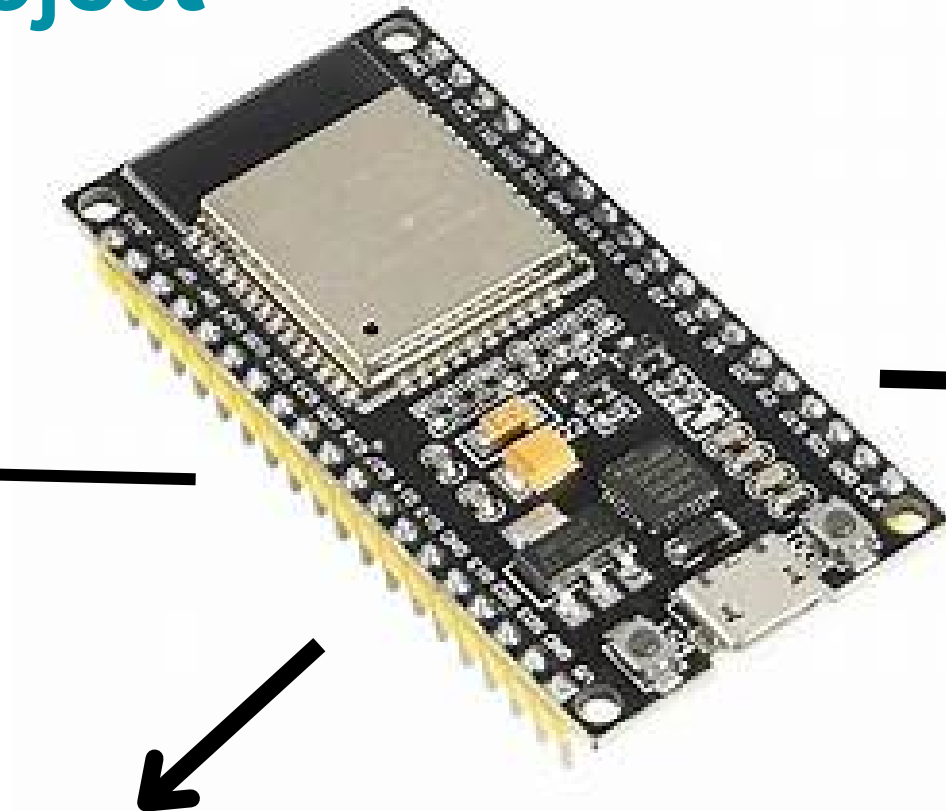


Home Automation

Controlled Devices in our project



Door Lock



Fan



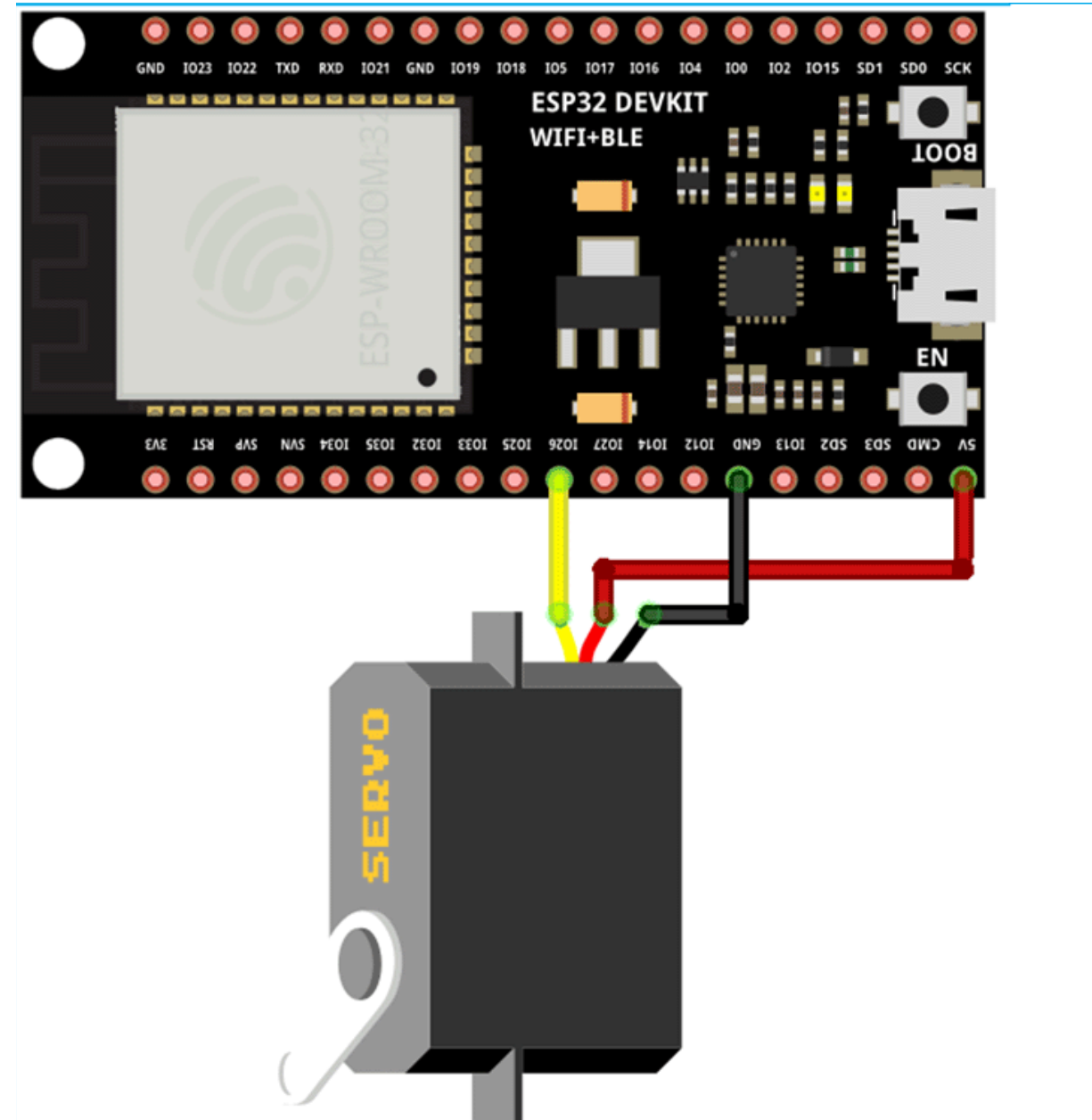
Bulb

Controlling Doorlock

Servo - Esp32 vcc

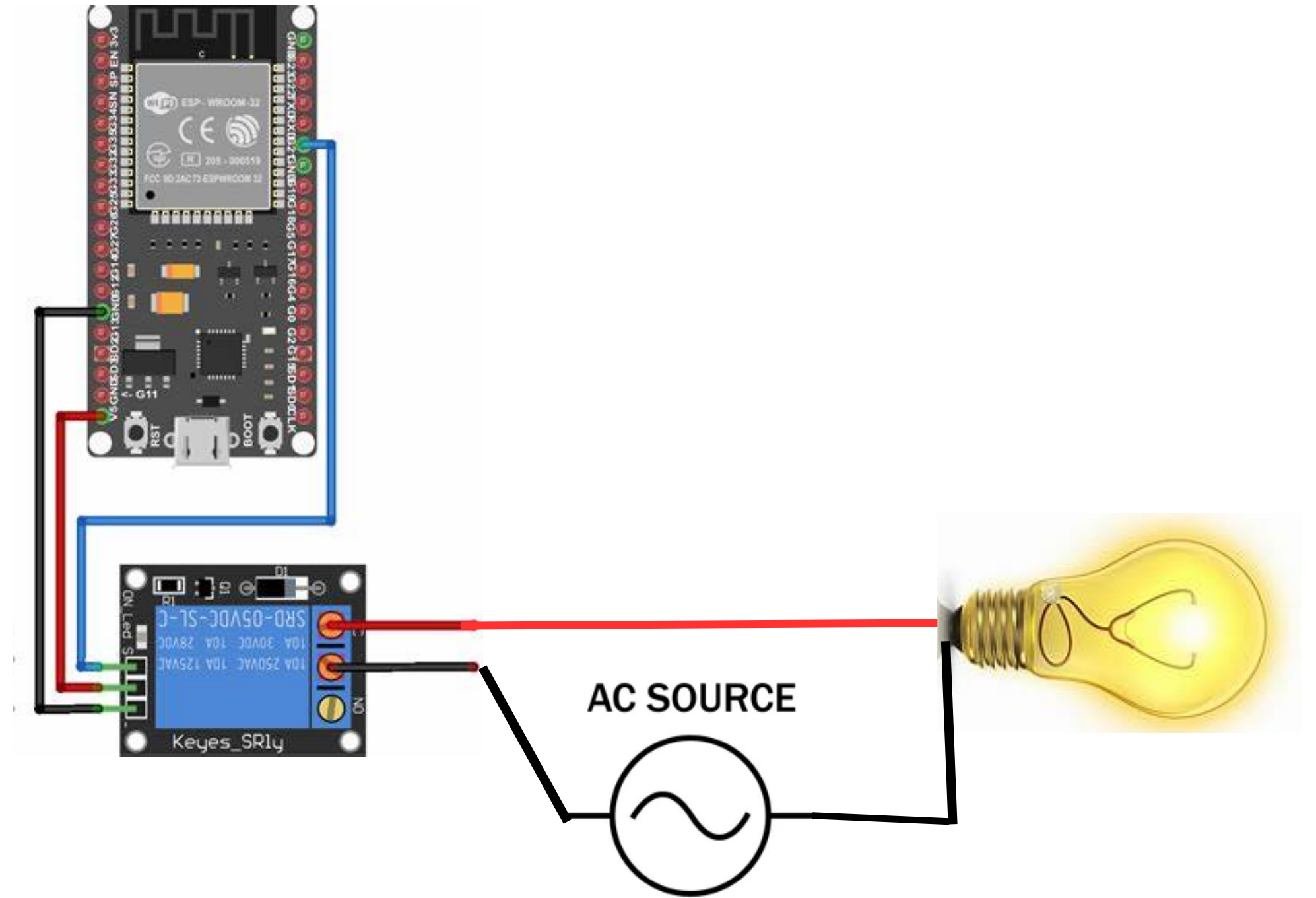
Servo Gnd -Esp32 Gnd

Servo digitalpin - digitalpin of
esp32



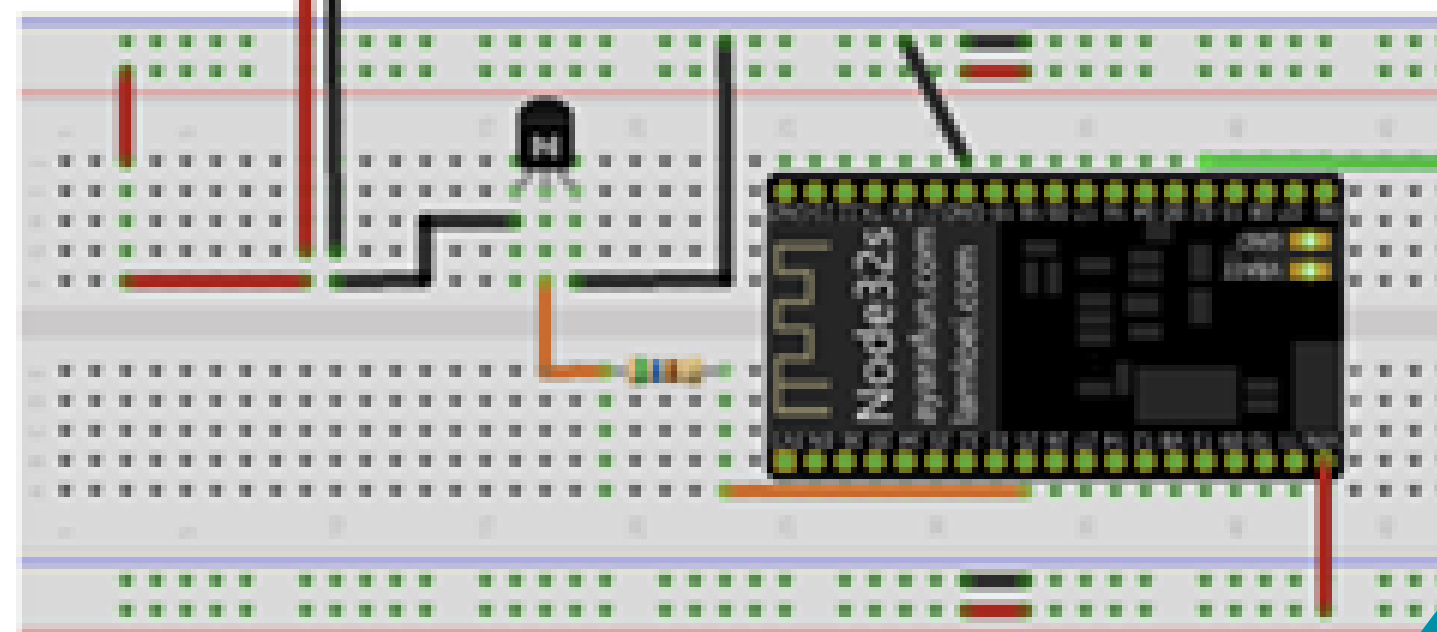
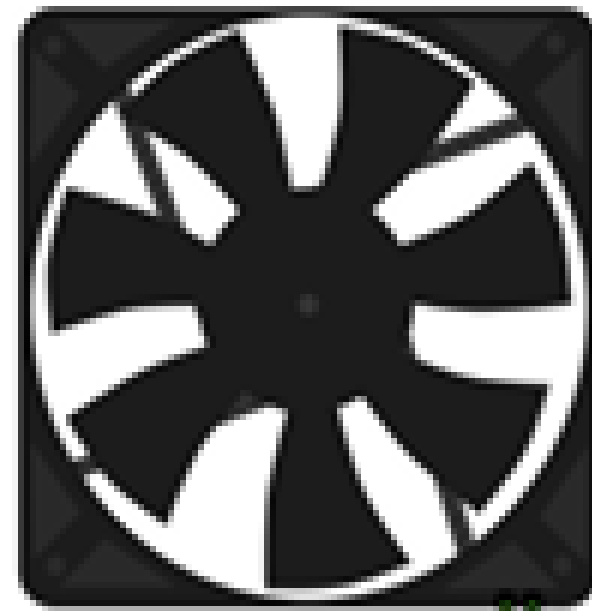
Controlling Bulb

1. relay Gnd - Gnd-Esp32
2. relay vcc - Esp32 vcc
3. relay input-esp32-digital pin



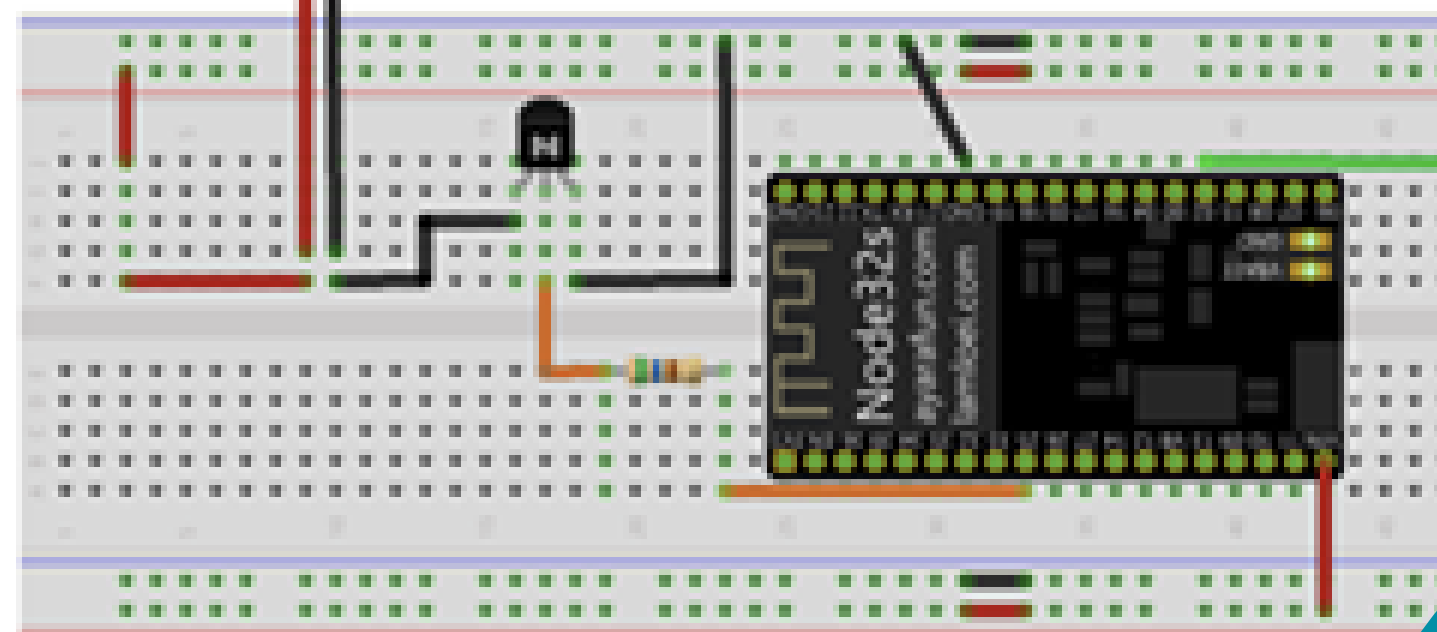
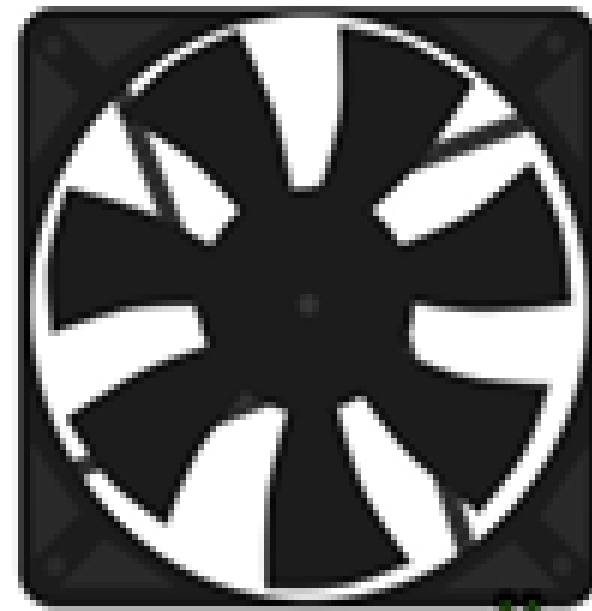
Fan Control

- 1.relay Gnd -Gnd-Esp32
- 2.relay vcc - Esp32 vcc
- 3.relay input-esp32-digital pin

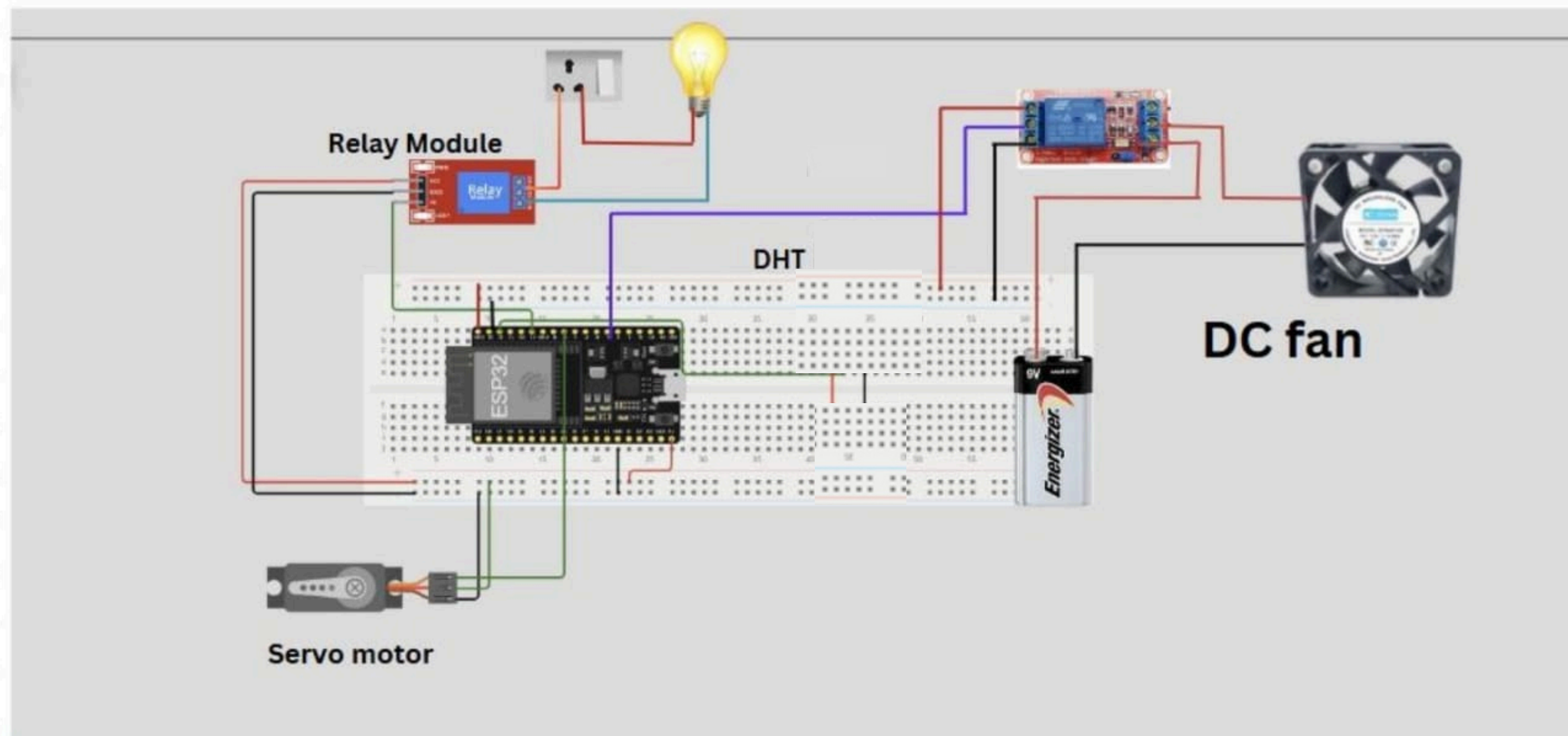


Fan Control

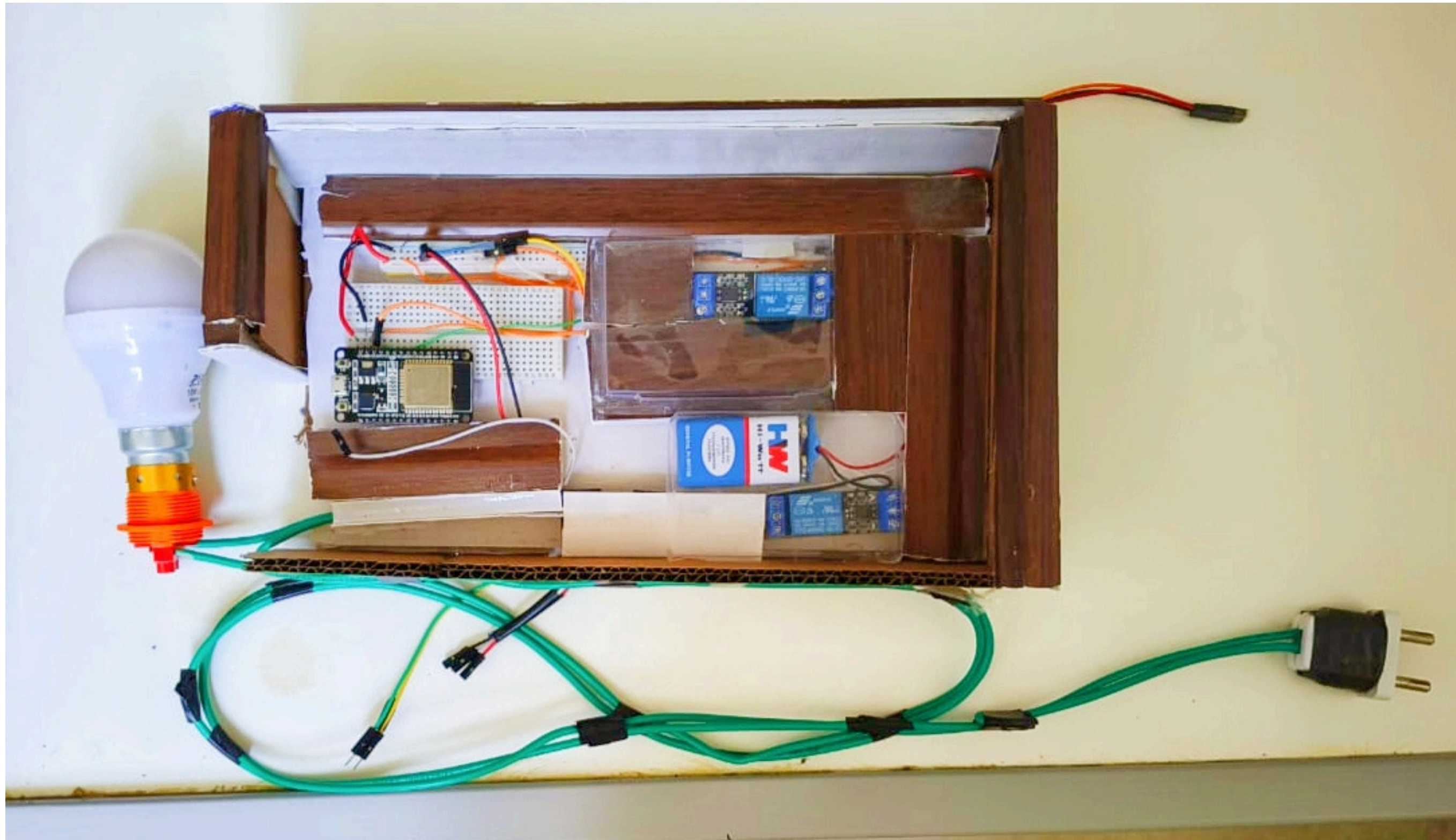
- 1.relay Gnd -Gnd-Esp32
- 2.relay vcc - Esp32 vcc
- 3.relay input-esp32-digital pin



A complete Home Automation setup Circuit Diagram

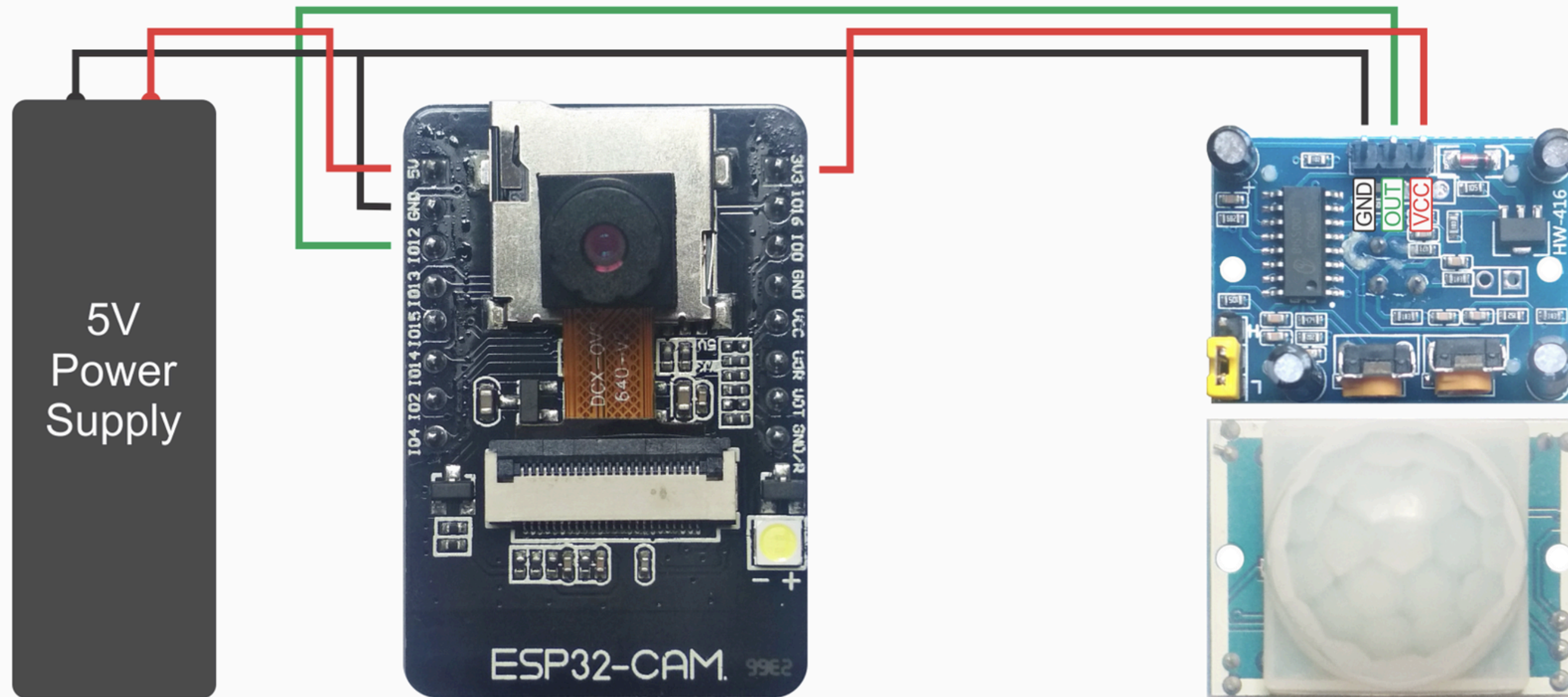


Home Automation Board



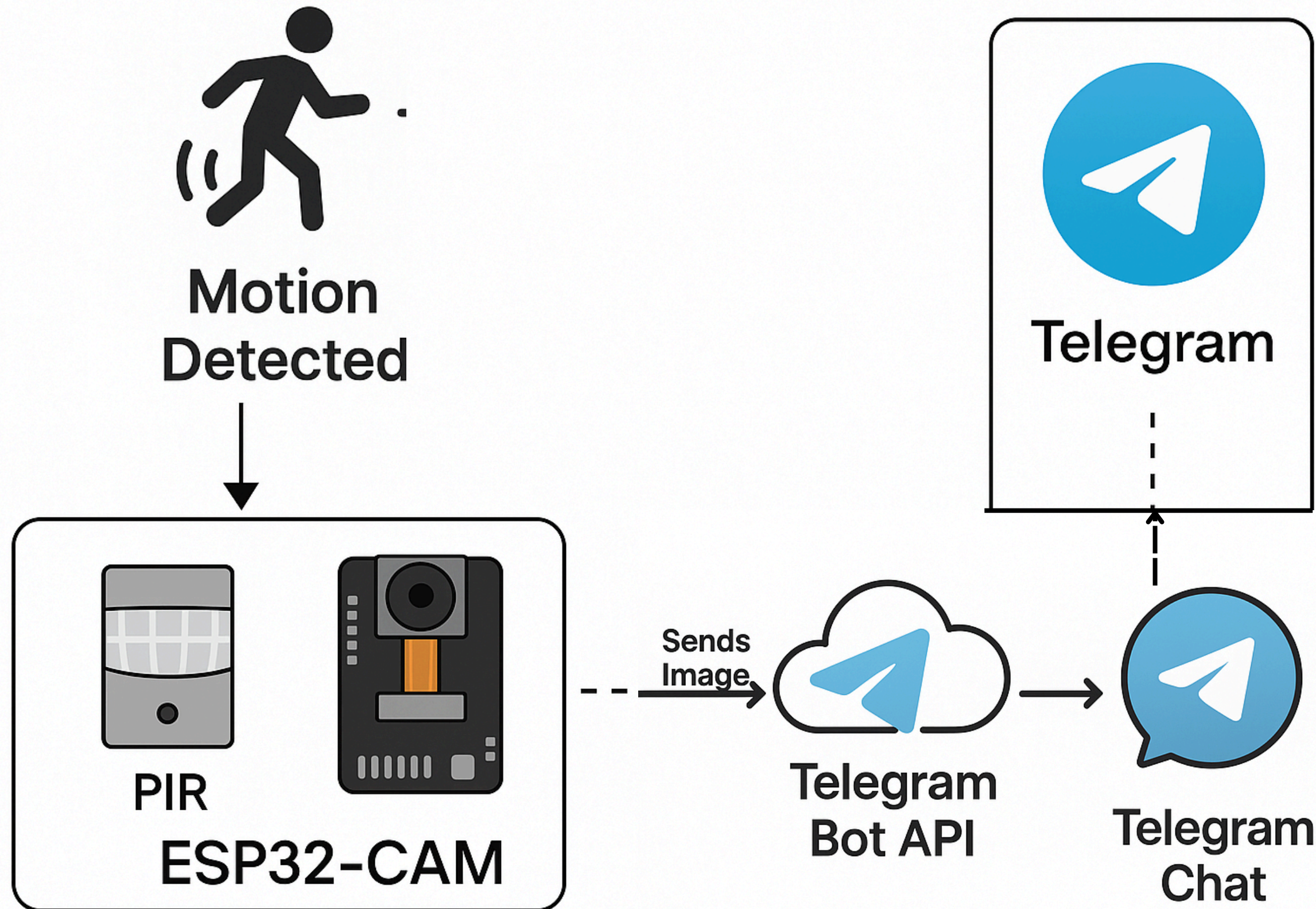
Motion detection cam

INSTALLATION



Motion detection cam

Working



Home Automation

Libraries Used

1. **machine , PWM** – Used to control GPIO pins and PWM (Pulse Width Modulation) for devices like LEDs, buzzers, and servo motors.
2. **time** – Used to add delays, measure time intervals, and schedule actions in the code.
3. **network** – Used to connect the ESP32 to a Wi-Fi network for cloud communication.
4. **logging** – Used to display messages for debugging and tracking the status of the program.
5. **arduino_iot_cloud** - Used to connect the ESP32 to Arduino IoT Cloud, send/receive data, and sync variables with the cloud dashboard.
6. **urequests** – Used to make HTTP/HTTPS requests, like sending photos to the Telegram API over the internet.
7. **camera** – Used to initialize and control the ESP32-CAM's OV2640 camera, capture JPEG images, and set resolution/quality (requires custom firmware like

Home Automation

IOT Cloud Applications



KMESmart

THANK YOU

